AVERY H. LIU

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EDUCATION

Brentwood School, Los Angeles
Class of 2029 Los Angeles, CA
High School Sophomore. Currently enrolled in Honors Chemistry, Honors/AP Precalculus, English, World History II Honors, Honors Spanish, Honors Independent Study, and Digital Art. Coursework emphasizes quantitative reasoning, scientific analysis, and written communication.

RESEARCH & PROJECTS

Characterizing the Microbiome Associated with Plastics in Urban Runoff Environments September 2026 - Present
Independent research project (planning stage)
- Designing an experiment using high-throughput sequencing to characterize the microbiome associated with plastics in Ballona Creek and compare it to control substrates.

WoundCorder July 2025 - June 2026
Smartphone wound assessment engineering project
- Assisting in the development of an AI-enabled smartphone wound assessment application in collaboration with Mitchell H. Liu.
- Supporting workflows that integrate LiDAR-based 3D sensing and computer vision to estimate wound area and volume and enable structured wound characterization for clinical contexts.

Evaluating Microplastic Contamination in Mussels on the Southern and Central California Coast October 2025 - June 2026
Field research project
- Planned and executed a coast-wide survey of microplastic contamination in mussels collected from urban creeks, jetties, and piers, sampling over 25 sites from Imperial Beach Pier in San Diego to San Simeon Pier in San Luis Obispo.
- Identified runoff-associated hotspots and established a baseline dataset to support future polymer characterization and microbiome studies.

RuKaiKai October 2024 - March 2025
Large-language-model translation chatbot
- Developed RuKaiKai, an LLM-powered chatbot delivering Paiwan-English translation to support preservation of endangered Formosan languages.
- Curated linguistic data, designed structured prompts, and iteratively refined outputs to improve translation accuracy and accessibility for over 100,000 speakers.

SELECTED RESEARCH OUTPUTS

1. Liu, A. H., Suzuki, H., Suzuki, S., & Liu, M. H. (2026). DermLatBench: Repurposing wound and dermatology image databases for AI laterality benchmarking [Conference presentation]. Symposium on Advanced Wound Care (SAWC) Fall, Las Vegas, NV, United States. (Accepted)
2. Liu, M. H., & Liu, A. H. (2026). WoundCorder: Smartphone-based LiDAR wound volumetry with local, offline vision-language model support for wound assessment. SAWC Fall. (Accepted)

HONORS & AWARDS

Johns Hopkins Global Health Leadership Conference - Selected, Student Speaker Series 2026
Los Angeles County Science Fair - Honorable Mention, Ecology 2026
Los Angeles County Science Fair - Honorable Mention, Mathematics & Computer Science 2025
Center for Future Global Leaders Spirit of Service Contest - 3rd Place (National) 2025
Congressional App Challenge - Recognition, Rep. Ted Lieu (CA-36): WoundCorder 2025
Presidential AI Challenge - Recognition: Fix and Flip AI 2025
Empowering American Self-Reliance Through Responsible AI
World Series of Innovation Challenge - 2nd Place, MetLife Foundation Good Health and Well-Being Track 2024-2025
Belldegrun Center for Innovative Leadership - Applied Science Finalist, Brentwood School 2024-2025
Model United Nations - Best Delegate
EAGLEMUN - Chair, UNICEF Committee

EXTRACURRICULARS & SERVICE

West Los Angeles VA Japanese Garden Restoration 2025 - Present
Volunteer
Rockreation Adaptive Climbing Program 2025 - Present
Volunteer
National Council of Jewish Women 2025 - Present
Volunteer

Southern California Chinese Lawyers Association 2024 - Present
Volunteer
USA Climbing 2025
Volunteer
YMCA Palm-Culver 2025
Volunteer
ICEA Lunar New Year Celebration 2025
Volunteer

ATHLETICS

Golf
Varsity Team, Brentwood School

Rock Climbing

ADDITIONAL INFORMATION

Skills: LLM prompt engineering and evaluation, research documentation and data organization, public speaking and meeting organization (Model United Nations), Microsoft Office, Google Workspace

Languages: English (native fluency), Spanish (working fluency), Mandarin (basic fluency)